# Progressive Hybrid-Order Hypergraph Framework for Hyperspectral Image Classification

Bing Yang, Wenke Yu, Ziyang Guo, Weiwei Sun, *Senior Member, IEEE*, and Jiangtao Peng, *Senior Member, IEEE*

***Abstract*—Hypergraph neural networks (HGNNs) have garnered considerable attention in hyperspectral image (HSI) classification for their ability to model higher order nonlinear relationships. However, most existing HGNN-based HSI classification methods adopt simple hypergraph structures, which fail to fully leverage low-order and high-order information within HSIs. Moreover, these methods are constrained by rigid single-scale superpixel segmentations, which fail to fully capture rich spectral–spatial features of HSIs. To overcome these limitations, this article proposes a progressive hybrid-order hypergraph framework (PHHF) that establishes a graph-hypergraph collaborative learning paradigm to learn multiscale hybrid-order features across hierarchical graphs. Specifically, by synergistically modeling multiple structural information, we develop a progressive HGNN framework to enhance the extraction of hierarchical representations of HSIs. Second, at each level, a novel hybrid-order hyperedge generation strategy is designed to enhance spectral–spatial consistency across scales and overcome the limitations of existing hypergraph methods. Finally, a novel heterogeneous kernel-based convolution (HetConv) is introduced to enhance pixel-level feature extraction for the PHHF. Compared with conventional convolution, HetConv offers richer spatial–spectral details with lower computational cost. Extensive experiments on widely used HSI datasets demonstrate that our proposed method achieves state-of-the-art HSI classification performance.**

***Index Terms*—Hybrid-order hypergraph, hypergraph neural network (HGNN), hyperspectral image (HSI) classification.**

## I. Introduction

HYPERSPECTRAL image (HSI) classification is an important domain within geoscience and remote sensing research. HSI classification aims to accurately identify and distinguish various materials or land cover types on the Earth's surface. HSIs provide rich spectral information across hundreds of bands and offer detailed spatial information, enabling precise identification of ground objects. This capability makes HSI classification highly valuable in various fields, such as precision farming, environmental protection, and urban monitoring [1], [2], [3].

With the advancement of deep learning, many deep neural networks have been applied to HSI classification and achieve significant results [4], [5], [6]. In particular, convolutional neural networks (CNNs) have become representative networks with strong spatial feature extraction capabilities due to their advantages, such as weight sharing and translation invariance [7], [8], [9]. For instance, Yu et al. [10] proposed a 2-D CNN model to extract valuable features for accurate HSI classification. Zhong et al. [11] developed a spectral–spatial residual network (SSRN) using stacked residual blocks to extract discriminative features. To overcome the limitations of rigid convolutional kernels, researchers have proposed adaptive convolution variants. For example, Zhu et al. [12] introduced deformable convolutions that utilized dynamic receptive fields. In [13], content-guided convolutions were adopted to adapt kernel shapes based on land cover distributions. Additionally, Tang et al. [14] proposed a 3-D octave convolution network with spatial–spectral attention to enhance discriminative feature learning in HSI classification. In [15], a lightweight multiscale attention fusion network was developed to integrate hierarchical features. While CNNs excel in local feature extraction for HSI classification, their fixed kernels may limit adaptability to diverse spatial scales and complex structures. Their localized receptive fields also make it challenging to capture long-range contextual relationships.

In recent years, graph neural networks (GNNs) have garnered significant attention due to their remarkable capability in long-range modeling of HSIs [16], [17], [18], [19]. Graph convolutional networks (GCNs) [20], a classic representative of GNNs, have been extensively utilized to learn the correlations between different land surfaces and to model their spatial topology using graph structures [21], [22], [23]. For instance, Mou et al. [24] introduced a non-local GCN to utilize relationships between HSI pixels. Qin et al. [25] designed a spectral–spatial GCN for HSI classification. However, treating each pixel as a node in the graph leads to significant computational demands and limits the effective utilization of spatial information. To address this challenge, rather than using individual pixels, several methods have opted to construct graph

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structures based on superpixels. For example, Wan et al. [26] adopted superpixels instead of individual pixels as nodes in the graph, which allows GCNs to effectively perform long-range modeling of HSIs. Bai et al. [27] assigned extracted superpixel features to original pixels with varying weights to enhance details. Yang et al. [28] highlighted the importance of individual spectra and conducted HSI classification within a global-to-local framework to better leverage contextual information. In [29], the CNN-enhanced GCN (CEGCN) was introduced to integrate the strengths of both CNNs and GCNs, leveraging locally generated information from CNNs to complement the superpixel-level information captured by GCNs. Zhou et al. [30] proposed an attention multihop graph and multiscale convolution fusion network (AMGCFN) to integrate multiscale CNN with multihop GCN. However, these approaches are unable to fully exploit the rich spatial structure information of HSIs, as they are limited to using only single-scale graphs. To tackle this problem, Guo et al. [31] proposed a dual graph U-Net architecture to effectively integrate spectral and spatial information of HSIs. Liu et al. [32] developed hierarchical superpixel convolutions to learn multiscale features. Qu et al. [33] enhanced HSI classification by integrating spectral–spatial attention into graph U-Net. Chen et al. [34] built an offset graph U-Net to extract features from HSIs and improve classification performance. Hu et al. [35] proposed a self-supervised graph masked autoencoders (SGMAEs) to extract multiscale spatial–spectral features from HSI. However, the aforementioned approaches fail to adequately model higher order interactions between pixels, limiting their ability to fully exploit the rich structural information inherent in HSIs.

Conventional GNN-based methods predominantly focus on graphs characterized by nodes linked through binary edges, which inherently overlook higher order relationships between nodes. In reality, the interactions among nodes extend beyond immediate neighbors, involving complex patterns that span multiple nodes. Hypergraph neural networks (HGNNs), by contrast, excel in capturing these higher order relationships among pixels, providing a richer and more flexible representation of HSIs [36]. For instance, in [37], a spectral–spatial hypergraph convolutional network was proposed to capture complex higher order and long-range correlations. Ma et al. [38] proposed a feature fusion HGNN that constructed hyperedges by leveraging multiple features extracted from HSI using the $k$-nearest neighbors (kNN) method. Duan et al. [39] proposed a structure-preserved hypergraph convolutional network that designed hypergraph convolutions under the guidance of high-order structures. Xu et al. [40] developed a hypergraph convolutional network that used superpixels as nodes and constructed hyperedges based on spectral similarity. Wang et al. [41] proposed a dual-branch model that combines an HGNN with hyperedge fusion and a multiscale CNN to enhance classification performance. Cui et al. [42] proposed a CNN-enhanced hypergraph attention (CEHGAT) to capture the dynamic relationships among hyperedges by leveraging an attention mechanism.

Despite recent advancements in HGNNs for HSI classification, existing methods face fundamental limitations in modeling complex intrinsic structures of HSIs. First, conventional hyperedge construction approaches, such as the kNN-based approach, rely on inflexible neighborhood mechanisms and fixed connectivity scales, which can hinder the ability to fully utilize rich spatial information in HSIs. Second, these methods also struggle to effectively establish complementary interactions between low-order and high-order relationships, leading to underutilization of spectral–spatial relationships. Moreover, existing HGNN-based methods typically operate on single-scale feature representations extracted from fixed-resolution superpixels, failing to fully exploit the multilevel spatial structures inherent in HSI. Consequently, there is an urgent need to explore more advanced and effective HGNN models to advance HSI classification.

To address these limitations, this article proposes a progressive hybrid-order hypergraph framework (PHHF) that establishes a graph-hypergraph collaborative learning paradigm through three fundamental innovations. First, we construct multiscale graph hierarchies via progressive superpixel merging, enabling simultaneous learning of spectral–spatial features across distinct spatial granularities. This hierarchical architecture captures both fine-grained details and high-level patterns through the hierarchical aggregation stage and progressive refinement stage. Second, we propose a novel hybrid-order hyperedge generation strategy that integrates three synergistic types. Specifically, the pairwise hyperedges are designed to preserve local geometric fidelity through spatial adjacency, the $k$-hop hyperedges are formulated to model long-range contextual dependencies with fixed neighborhood hops, and the merging hyperedges are proposed to encode cross-scale superpixel aggregation patterns. This strategy establishes comprehensive low-to-high order connectivity while overcoming the limitations of conventional kNN-based approaches. Finally, we utilize the heterogeneous kernel-based convolution (HetConv) to extract pixel-level features, which employs larger kernels and smaller pointwise kernels within the same filter. This heterogeneous kernel fusion enhances spectral–spatial feature representation while maintaining computational efficiency. The main contributions are summarized as follows.

1) A novel PHHF is proposed for HSI classification by integrating graph learning and hypergraph learning into a unified progressive hypergraph framework to extract hierarchical hybrid-order features and improve classification performance. The PHHF demonstrates its superiority compared with state-of-the-art HSI classification methods.
2) A progressive HGNN framework is designed to fully mine multiscale contextual features through hierarchical hypergraph propagation. Besides, an innovative hybrid-order hyperedge generation strategy is built to incorporate hyperedges of various types to extract complementary spatial information.
3) The HetConv is introduced to enhance pixel-level details, complementing the superpixel-level information obtained from the progressive HGNN framework. This effectively bridges fine-grained pixel-level patterns with superpixel-level patterns.

The remainder of this article is organized as follows. The proposed PHHF is introduced in Section II. The comparative experimental results and analysis are exhibited in Section III. The conclusions are presented in Section IV.

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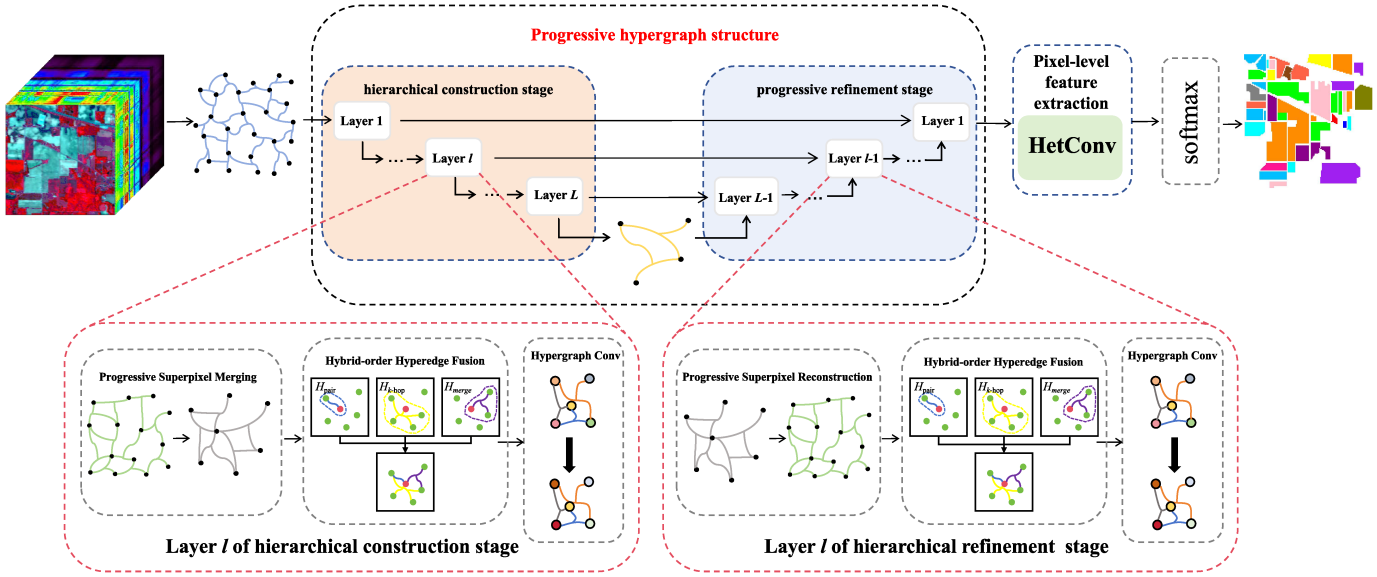



Fig. 1. Overall framework of the proposed PHHF. First, the HSI is segmented into superpixels at multiple scales. Then, a progressive hypergraph framework is employed to extract hybrid-order spectral–spatial features, followed by HetConv for pixel-level feature refinement. Finally, a classifier is applied to generate the classification result.

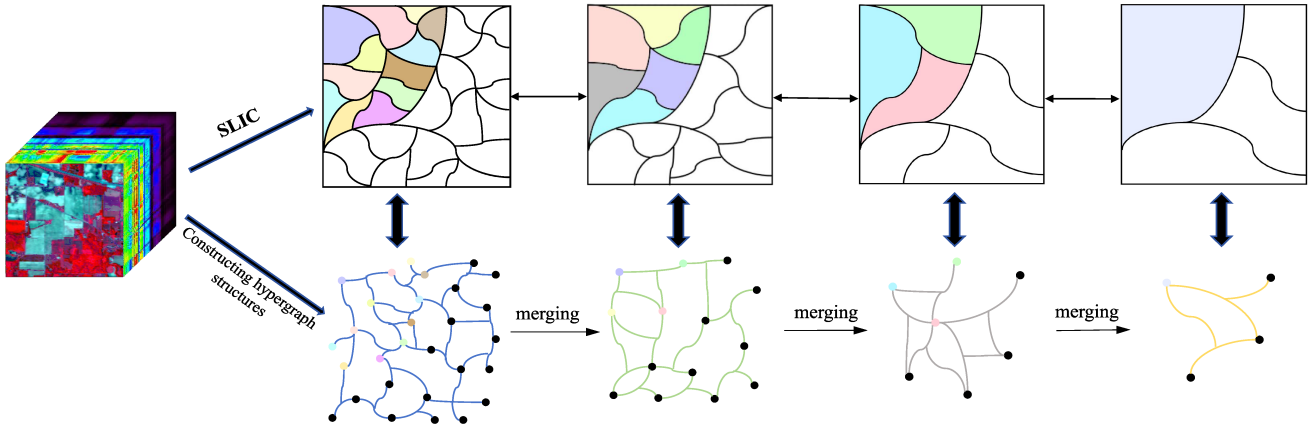



Fig. 2. Process of constructing hierarchical hypergraph structure.

## II. Proposed Method

In this section, we provide a detailed introduction to the proposed PHHF. The complete architecture of our model is illustrated in Fig. 1. First, the hyperspectral data is fed into the progressive framework, where the HSI is segmented into superpixels at different scales to extract spectral–spatial information from superpixels of varying sizes. During the hierarchical construction stage, we design a progressive hypergraph structure (PHS) and utilize HGNN to build hybrid-order hyperedges at different scales. In the progressive refinement stage, we integrate the spatial–spectral information extracted during the hierarchical construction stage to refine feature extraction. Finally, the extracted superpixel features are fed into the HetConv module and the classifier for classification.

### A. Progressive Superpixel Segmentation

In this work, the HSI is denoted as $\boldsymbol{X} \in \mathbb{R}^{H\times W\times B}$, where $H$, $W$, and $B$ represent its height, width, and the number of spectral bands, respectively. Each pixel is represented by $\boldsymbol{x}_i \in \mathbb{R}^B$, with $i = 1, 2, \ldots, n$ and $n = HW$. To fully model spatial topologies of HSIs, we propose to convert HSIs into a series of hierarchical superpixel segmentations and represent them by hypergraphs. Specifically, as illustrated in Fig. 2, the HSI is initially segmented into numerous superpixels and represented by an initial hypergraph. Then, the initial superpixels are progressively merged according to spatially adjacent relations to generate hypergraphs with progressive hierarchies. The details are given as follows.

First, we begin by segmenting the input HSI into a set of first-level superpixels $\mathcal{S}^{(1)} = \{s_i^{(1)}\}_{i=1}^{m^{(1)}}$, where $m^{(1)}$ is the total number of superpixels at this level. By treating each superpixel $s_i^{(1)}$ as a node $v_i^{(1)}$, we define a graph $\mathcal{G}^{(1)} = (\mathcal{V}^{(1)}, \mathcal{E}^{(1)})$ with node set $\mathcal{V}^{(1)}$ and edge set $\mathcal{E}^{(1)}$. These nodes are represented by a feature matrix $\boldsymbol{F}^{(1)} \in \mathbb{R}^{m^{(1)}\times D}$, where $m^{(1)}$ is the number of nodes (superpixels) at level 1, and $D$ is the dimensionality of the features. The edges $\mathcal{E}^{(1)}$ is represented by an adjacency matrix $\boldsymbol{A}^{(1)}$, which encodes the spatial adjacency relationships

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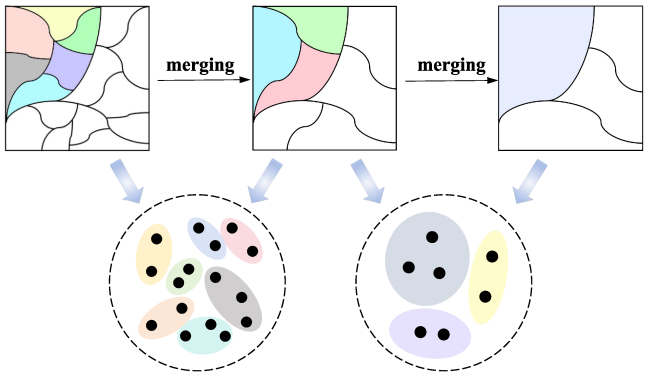



Fig. 3. Illustration of the formation process of merging hyperedges.

between superpixels. Furthermore, the hyperedges are generated using the approach described in Section II-B.

Second, the Boruvka algorithm [43] is employed to progressively merge adjacent superpixels within $\mathcal{S}^{(1)}$, resulting in a coarser second-level superpixel set $\mathcal{S}^{(2)} = \{s_i^{(2)}\}_{i=1}^{m^{(2)}}$, where $m^{(2)} < m^{(1)}$. Similarly, we can define a corresponding graph $\mathcal{G}^{(2)} = (\mathcal{V}^{(2)}, \mathcal{E}^{(2)})$, and the feature matrix and the adjacency matrix are represented by $\boldsymbol{F}^{(2)}$ and $\boldsymbol{A}^{(2)}$, respectively.

Finally, by iteratively applying the merging process, a series of increasingly coarser superpixel sets can be generated. Let the process conclude at the $L$th level, then we can obtain hierarchical segmentations $\{\mathcal{S}^{(l)}\}_{l=1}^{L}$ and corresponding graphs $\{\mathcal{G}^{(l)} = (\mathcal{V}^{(l)}, \mathcal{E}^{(l)})\}_{l=1}^{L}$. These processes yield a hierarchical representation of the HSI.

To flexibly transfer features between any two consecutive levels of graphs, we give the following propagation rules. Specifically, let $\boldsymbol{Q}^{(l)} \in \mathbb{R}^{|\mathcal{S}^{(l-1)}| \times |\mathcal{S}^{(l)}|}$ $(2 \le l \le L)$ be the association matrix between $\mathcal{S}^{(l-1)}$ and $\mathcal{S}^{(l)}$, which is calculated by

$$Q_{i,j}^{(l)} = \begin{cases} 1, & \text{if } s_i^{(l-1)} \subseteq s_j^{(l)} \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

Based on the association matrix $\boldsymbol{Q}^{(l)}$, we establish feature propagation rules across hierarchical segmentation levels

$$\begin{cases} \boldsymbol{F}^{(l)} = \left(\tilde{\boldsymbol{Q}}^{(l)}\right)^{\top} \boldsymbol{F}^{(l-1)} \\ \tilde{\boldsymbol{F}}^{(l-1)} = \boldsymbol{Q}^{(l)} \boldsymbol{F}^{(l)}. \end{cases} \tag{2}$$

Here, $\tilde{\boldsymbol{Q}}^{(l)}$ represents the normalized association matrix obtained by normalizing along columns, where $\tilde{Q}_{i,j}^{(l)} = Q_{i,j}^{(l)} / \sum_k Q_{k,j}^{(l)}$. Particularly, let $\boldsymbol{Q}^{(1)} \in \mathbb{R}^{HW \times m^{(1)}}$ denote the association matrix between HSI pixels and $\mathcal{S}^{(1)}$.

### B. Hybrid-Order Hypergraph Structure

The construction of hyperedges critically determines hypergraphs' capacity to model complex relationships in HSIs. Considering the limitations in conventional kNN-based methods, we develop three specialized hyperedge types and combine them through the hybrid generation strategy.

*1) Construct Merging Hyperedges:* In this method, merging hyperedges are constructed by merging spatially adjacent and spectrally similar superpixels, as shown in Fig. 3. Specifically, when a group of superpixels in the current layer merges into a single superpixel at the next hierarchical level, they form a unique merging hyperedge that represents their merging relationship. Thus, the merging hyperedges can encode cross-scale superpixel aggregation patterns.

To achieve hierarchical feature fusion, we propose a hyperedge generation method based on hierarchical inclusion relationships. Here, each superpixel is treated as a vertex in the hypergraph.Given the set of superpixels $\mathcal{S}^{(l)} = \{s_1^{(l)}, s_2^{(l)}, \ldots, s_{m^{(l)}}^{(l)}\}$ at level $l$, the corresponding vertex set is defined as $\mathcal{V}^{(l)} = \{v_1^{(l)}, v_2^{(l)}, \ldots, v_{m^{(l)}}^{(l)}\}$. For simplicity, we use the terms "superpixel" and "vertex" interchangeably.

When transitioning from level $l$ to a coarser level $l+1$, each high-level superpixel $s_v^{(l+1)}$ is formed by merging several low-level superpixels, and the inclusion relationship is defined as

$$\mathcal{N}(v^{(l+1)}) = \left\{ v^{(l)} \mid s_{v^{(l)}}^{(l)} \subseteq s_{v^{(l+1)}}^{(l+1)},\ v^{(l)} \in \mathcal{V}^{(l)} \right\} \tag{3}$$

where $s_{v^{(l)}}^{(l)}$ is the superpixel corresponding to the vertex $v^{(l)}$ at level $l$, and $s_{v^{(l+1)}}^{(l+1)}$ is the superpixel corresponding to the vertex $v^{(l+1)}$ at level $l+1$. The merging hyperedge set at level $l$ is then generated as

$$\boldsymbol{\varepsilon}_{\text{merge}}^{(l)} = \left\{ e \triangleq \mathcal{N}(v^{(l+1)}) \mid\ |\mathcal{N}(v^{(l+1)})| \ge 2,\ v^{(l+1)} \in \mathcal{V}^{(l+1)} \right\}. \tag{4}$$

The condition $|\mathcal{N}(v^{(l+1)})| \ge 2$ eliminates redundant hyperedges, ensuring only meaningful merging relationships are represented.

The corresponding hyperedge incidence matrix $\boldsymbol{H}_{\text{merge}}^{(l)} \in \{0,1\}^{m^{(l)} \times m^{(l+1)}}$ is constructed as follows:

$$(\boldsymbol{H}_{\text{merge}}^{(l)})_{ij} = \begin{cases} 1, & \text{if } v_i \in e_j \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

where $e_j \in \boldsymbol{\varepsilon}_{\text{merge}}^{(l)}$.

This construction method ensures that the resulting merging hypergraphs effectively capture spatial and hierarchical relationships between superpixels. By encoding the merging process into hyperedges, it not only preserves local structural information but also supports multiscale analysis.

*2) Construct k-hop Hyperedges:* The second strategy focuses on generating $k$-hop high-order hyperedges $\mathcal{E}_{\text{hop}}$, where each hyperedge connects a central node to all its $k$-hop neighbors. For example, 1-hop refers to the direct neighbors of the central node, two-hop refers to the neighbors of the central node's neighbors, and so on. This approach extends the search radius in the graph structure to capture broader relationships. The specific generation method of $k$-hop hyperedges is given as follows.

Let $\boldsymbol{A}$ denote the adjacency matrix of the graph $\mathcal{G}^{(l)}$, then the $k$-hop neighborhood of a vertex $v$ is defined as

$$N_{\text{hop}_k}(v) = \left\{ u \mid (\boldsymbol{A}^k)_{uv} \ne 0, u \in \mathcal{V}^{(l)} \right\} \tag{6}$$

where $\boldsymbol{A}^k$ represents the $k$th power of the adjacency matrix, capturing all nodes reachable within $k$ hops from $v$.

Using this definition, the hyperedge group with $k$-hop neighbors can be expressed as

$$\mathcal{E}_{\text{hop}_k}^{(l)} = \left\{ \tilde{e} \triangleq N_{\text{hop}_k}(v) \mid v \in \mathcal{V}^{(l)} \right\}. \tag{7}$$

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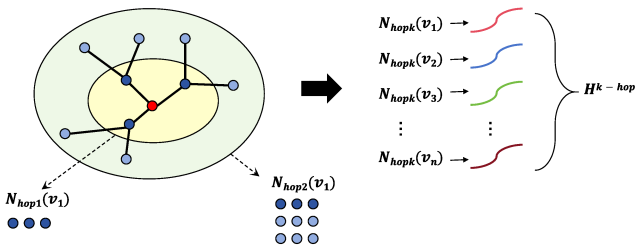



Fig. 4. Illustration of $k$-hop hyperedge construction.

Each hyperedge in $\mathcal{E}^{(l)}_{\text{hop}_k}$ includes a central node $v$ and all its $k$-hop neighbors.

To encode these hyperedges, we construct the $k$-hop hyperedge matrix $\boldsymbol{H}^{(l)}_{k\text{-hop}} \in \{0,1\}^{m^{(l)} \times m^{(l)}}$. The elements of $\boldsymbol{H}^{(l)}_{k\text{-hop}}$ are defined as

$$(\boldsymbol{H}^{(l)}_{k\text{-hop}})_{ij} = \begin{cases} 1, & \text{if } v_i \in \tilde{e}_j \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

where $\tilde{e}_j \in \mathcal{E}^{(l)}_{\text{hop}_k}$. This formulation enables $\mathcal{E}^{(l)}_{\text{hop}_k}$ to exploit external correlated vertices for the central node by extending the search radius in the graph structure.

As shown in Fig. 4, this approach captures a broader range of neighbors and provides richer correlation information essential for modeling higher order relationships.

*3) Construct Pairwise Hyperedges:* While capturing high-order information is crucial for modeling complex hyperspectral patterns, we explicitly preserve fundamental low-order spatial relationships through pairwise hyperedges. These two-uniform hyperedges establish direct connections between spatially adjacent superpixels, which are formally defined as

$$\boldsymbol{\varepsilon}^{(l)}_{\text{pair}} = \left\{ \bar{e} \triangleq \{v_i, v_j\} \mid (v_i, v_j) \in \mathcal{E}^{(l)} \right\} \tag{9}$$

where $\mathcal{E}^{(l)}$ denotes the edge set of the graph $\mathcal{G}^{(l)}$ at level $l$, with adjacency determined by spatial bordering between superpixels $s^{(l)}_i$ and $s^{(l)}_j$.

The corresponding hyperedge incidence matrix $\boldsymbol{H}_{\text{pair}} \in \{0,1\}^{m^{(l)} \times |\mathcal{E}^{(l)}|}$ is constructed as

$$(\boldsymbol{H}^{(l)}_{\text{pair}})_{ij} = \begin{cases} 1, & \text{if } v_i \in \bar{e}_j \\ 0, & \text{otherwise} \end{cases} \tag{10}$$

where $\bar{e}_j \in \boldsymbol{\varepsilon}^{(l)}_{\text{pair}}$. This construction preserves the geometric structure of superpixel decompositions through direct mapping of spatial adjacencies. The resulting low-order correlations provide a foundation for hierarchical feature integration in subsequent hypergraph processing stages.

The hybrid-order hyperedge set combines three components

$$\boldsymbol{H}^{(l)} = \boldsymbol{H}^{(l)}_{\text{merge}} \parallel \boldsymbol{H}^{(l)}_{k\text{-hop}} \parallel \boldsymbol{H}^{(l)}_{\text{pair}} \tag{11}$$

where $\parallel$ denotes concatenation. The fused hyperedge matrix integrates hierarchical spatial–spectral information, enabling robust feature extraction through progressive hypergraph convolutions that capture complex data relationships.

### C. Progressive Hypergraph Feature Extraction

Existing HGNN-based methods are limited to single-level modeling, failing to fully exploit multiscale spatial relationships in HSIs. The lack of multiscale analysis may lead to insufficient use of contextual information, especially for land covers with varying sizes and spectral heterogeneity. To this end, we develop a progressive HGNN framework, which is detailed as follows.

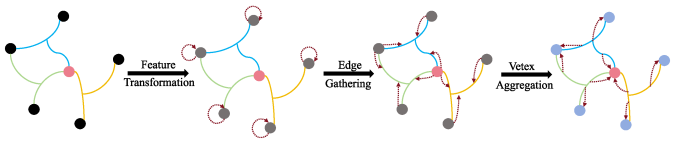



Fig. 5. Propagation process of hypergraph convolution.

First, we introduce the hypergraph convolution. Fig. 5 shows how vertex features are aggregated to hyperedges and then redistributed back to vertices in the hypergraph convolution.

Mathematically, given the hypergraph incidence matrix $\boldsymbol{H}$ and the node feature matrix $\boldsymbol{F}$, a single-layer hypergraph convolution $f_{\mathcal{HG}}$ can be formulated as

$$f_{\mathcal{HG}}(\boldsymbol{H}, \boldsymbol{F}) \triangleq \sigma\left(\boldsymbol{D}_v^{-\frac{1}{2}} \boldsymbol{H} \boldsymbol{W}_{\text{HG}} \boldsymbol{D}_e^{-1} \boldsymbol{H}^T \boldsymbol{D}_v^{-\frac{1}{2}} \boldsymbol{F} \boldsymbol{\Theta}\right) \tag{12}$$

where $\boldsymbol{\Theta}$ is the learnable weight matrix, $\boldsymbol{W}_{\text{HG}}$ denotes the hyperedge weight matrix, $\sigma(\cdot)$ represents the LeakyReLU activation function, $\boldsymbol{D}_v$ signifies the degree matrix for vertices, and $\boldsymbol{D}_e$ is the diagonal matrix representing hyperedge degrees.

Subsequently, based on the hypergraph convolution $f_{\mathcal{HG}}$, we build the progressive HGNN framework to leverage the spatial topology of HSI. Specifically, as shown in Fig. 1, the HSI first passes through a fully connected layer, and then is fed into the hierarchical construction stage, where features are transferred to different levels of graphs. Assuming that the PHHF contains $L$ layers in the hierarchical construction stage, then the feature extraction process can be represented as follows:

$$\begin{cases} \tilde{\boldsymbol{F}}^{(l)} = \text{BN}\left(\left(\boldsymbol{Q}^{(l)}\right)^{\top} \boldsymbol{F}^{(l-1)}\right) \\ \boldsymbol{F}^{(l)} = f_{\mathcal{HG}}\left(\boldsymbol{H}^{(l)}, \tilde{\boldsymbol{F}}^{(l)}\right) \end{cases} \tag{13}$$

where $l = 1, 2, \ldots, L$, $\text{BN}(\cdot)$ denotes the batch normalization. Additionally, $\boldsymbol{F}^{(l)}$ is the output feature map of the $l$th hierarchical construction stage. $\boldsymbol{F}^{(0)} = \hat{\boldsymbol{X}} \boldsymbol{W}_{\text{pre}}$, where $\hat{\boldsymbol{X}} = \text{flatten}(\boldsymbol{X}) \in \mathbb{R}^{HW \times B}$ denotes the flattened HSI, $\boldsymbol{W}_{\text{pre}} \in \mathbb{R}^{B \times D}$ is a learnable weight matrix, and $D$ denotes the number of neurons in the hypergraph convolution. By this means, the hierarchical construction stages can learn hierarchical features across different levels of graphs, thereby more comprehensively exploiting the spatial structures within HSI.

In the progressive refinement stage, we opt for a coarse-to-fine progressive fusion of hierarchical features from the hierarchical construction stages, rather than directly concatenating features of different scales. Let $\boldsymbol{Z}^{(l)}$ denote the fused feature map of the $l$th progressive refinement stage; we have

$$\begin{cases} \tilde{\boldsymbol{Z}}^{(l)} = \boldsymbol{Q}^{(l+1)} \boldsymbol{Z}^{(l+1)} \\ \hat{\boldsymbol{Z}}^{(l)} = \text{BN}\left(\text{concat}\left(\tilde{\boldsymbol{Z}}^{(l)}, \boldsymbol{F}^{(l)}\right)\right) \\ \boldsymbol{Z}^{(l)} = f_{\mathcal{HG}}\left(\boldsymbol{H}^{(l)}, \hat{\boldsymbol{Z}}^{(l)}\right) \end{cases} \tag{14}$$

where $l = L-1, L-2, \ldots, 1$, and $\text{concat}(\cdot)$ denotes concatenating features along the channel dimension. Particularly, $\boldsymbol{Z}^{(L)} = \boldsymbol{F}^{(L)}$. The progressive refinement stage progressively fuses the hierarchical features learned by the hierarchical

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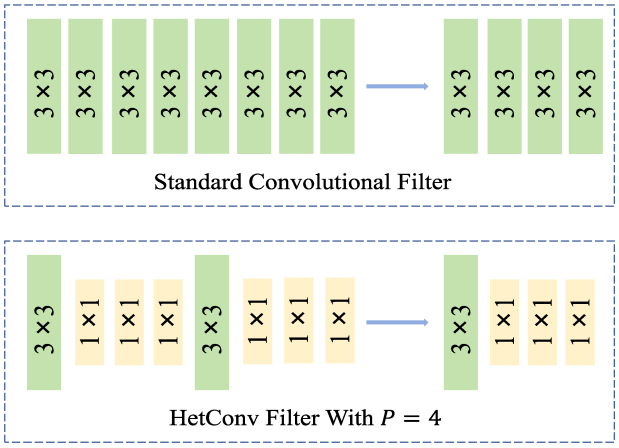



Fig. 6. Intuitive comparison between standard convolution and the HetConv.

construction stage, resulting in the final superpixel-level representations. Compared with directly concatenating features of different scales, the progressive fusion mechanism allows for more refined feature extraction and better preservation of spatial–spectral coherence.

The extracted superpixel-level features $\boldsymbol{Z}^{(L)}$ is then concatenate with the pixel-level features $\boldsymbol{F}^{(0)}$ and fed into the HetConv as

$$\boldsymbol{X}_{\mathrm{HC}} = \mathrm{concat}\left(\boldsymbol{F}^{(0)}, \boldsymbol{Q}^{(1)}\boldsymbol{Z}^{(1)}\right) \tag{15}$$

where $\boldsymbol{Q}^{(1)}$ represents the association matrix between the pixel-level and superpixel features defined in Section II-A.

### D. HetConv

HSI classification aims to assign labels to pixels, rather than superpixels. Therefore, after obtaining the superpixel-level features through the PHHF, we further introduce the HetConv [44] to enhance the representation by incorporating more refined pixel-level features.

As shown in Fig. 6, different from the standard convolution, the HetConv utilizes heterogeneous kernels within a single filter, whereby each filter applies convolutional kernels of different sizes across distinct subsets of input channels. Specifically, the input feature $\boldsymbol{X}_{\mathrm{HC}}$ is first divided into $D/P$ groups, and each group contains $P$ channels, where $P$ is a hyperparameter (suppose that $D$ can be divisible by $P$). Then, the first channel in each group, totaling $D/P$ channels, is convolved with $K \times K$ kernels, whereas the remaining channels are convolved with $1 \times 1$ kernels. The formula of the HetConv is given as

$$\begin{aligned}\boldsymbol{F}_{\mathrm{out}}(i,j) &= \sum_{m=1}^{D/P} (\boldsymbol{X}_{\mathrm{HC}}(i:i+K-1, j:j+K-1, m) \\ &\quad * \boldsymbol{W}_{KK}(m)) \\ &\quad + \sum_{n=D/P+1}^{D} (\boldsymbol{X}_{\mathrm{HC}}(i,j,n) \odot \boldsymbol{W}_{11}(n)) \\ &\triangleq f_{\mathcal{HC}}(\boldsymbol{X}_{\mathrm{HC}})\end{aligned} \tag{16}$$

where $*$ denotes the convolution operation and $\odot$ represents element-wise multiplication. Here, $\boldsymbol{W}_{KK}(m)$ denotes the $K \times K$ kernel applied to the $m$th channel in the corresponding group, while $\boldsymbol{W}_{11}(n)$ denotes the $1 \times 1$ kernel for the $n$th channel.

Compared with standard convolution, the reduction in computational cost achieved by the HetConv is

$$R_{\mathrm{HetConv}} = \frac{1}{P} + \frac{(1-1/P)}{K^2}. \tag{17}$$

As a result, our PHHF can not only fully utilize both spatial and channel-wise information, but also balance computational efficiency and model performance well.

### E. Loss Function and Classifier Design

To predict the class probabilities for each pixel, we employ a softmax classifier that operates on the feature $\boldsymbol{F}_{\mathrm{out}}$. Specifically, the softmax classifier is defined as

$$\boldsymbol{Y} = \mathrm{Softmax}(\boldsymbol{W}_{\mathrm{out}}\boldsymbol{F}_{\mathrm{out}} + \boldsymbol{b}_{\mathrm{out}}) \tag{18}$$

where $\boldsymbol{W}_{\mathrm{out}}$ and $\boldsymbol{b}_{\mathrm{out}}$ are the weight matrix and the bias of the classifier, respectively.

**Algorithm 1** PHHF

**Input:** HSI $\boldsymbol{X}$, association matrices $\{\boldsymbol{Q}^{(l)}\}_{l=1}^{L}$ between hierarchical graphs, the learning rate $\eta$, the number of training iterations $T_{\max}$.
**Output:** The trained parameters $\boldsymbol{W}$.

Set $t = 1$, and randomly initialize $\boldsymbol{W}$;
$\boldsymbol{F}^{(0)} = \hat{\boldsymbol{X}}\boldsymbol{W}_{\mathrm{pre}}$, where $\hat{\boldsymbol{X}}$ denotes the flattened HSI;
**repeat**
  **for** $l = 1, 2, \cdots, L$ **do**
    $\tilde{\boldsymbol{F}}^{(l)} = \mathrm{BN}\big((\hat{\boldsymbol{Q}}^{(l)})^{\top}\boldsymbol{F}^{(l-1)}\big)$;
    $\boldsymbol{H}^{(l)} = \boldsymbol{H}^{(l)}_{\mathrm{merge}} \,\|\, \boldsymbol{H}^{(l)}_{k\text{-hop}} \,\|\, \boldsymbol{H}^{(l)}_{\mathrm{pair}}$;
    $\mathbf{F}^{(l)} = f_{\mathcal{HG}}\big(\boldsymbol{H}^{(l)}, \tilde{\mathbf{F}}^{(l)}\big)$;
  **end**
  $\boldsymbol{Z}^{(L)} = \boldsymbol{F}^{(L)}$;
  **for** $l = L-1, L-2, \cdots, 1$ **do**
    $\overline{\boldsymbol{Z}}^{(l)} = \boldsymbol{Q}^{(l+1)}\boldsymbol{Z}^{(l+1)}$;
    $\widehat{\boldsymbol{Z}}^{(l)} = \mathrm{BN}\Big(\mathrm{concat}\Big(\overline{\boldsymbol{Z}}^{(l)}, \boldsymbol{F}^{(l)}\Big)\Big)$;
    $\boldsymbol{Z}^{(l)} = f_{\mathcal{HG}}\Big(\boldsymbol{H}^{(l)}, \widehat{\boldsymbol{Z}}^{(l)}\Big)$;
  **end**
  $\boldsymbol{X}_{\mathrm{HC}} = \mathrm{concat}\big(\boldsymbol{Q}^{(1)}\boldsymbol{Z}^{(1)}, \boldsymbol{X}\big)$;
  $\boldsymbol{F}_{\mathrm{out}} = f_{\mathcal{HC}}(\boldsymbol{X}_{\mathrm{HC}})$;
  $\boldsymbol{Y} = \mathrm{Softmax}(\boldsymbol{W}_{\mathrm{out}}\boldsymbol{F}_{\mathrm{out}} + \boldsymbol{b}_{\mathrm{out}})$;
  Calculate cross-entropy loss $\mathcal{L}_{\mathrm{PHHF}}(\boldsymbol{W})$;
  Update parameters: $\boldsymbol{W} \leftarrow \boldsymbol{W} - \eta\dfrac{\partial \mathcal{L}_{\mathrm{PHHF}}(\boldsymbol{W})}{\partial \boldsymbol{W}}$;
  $t \leftarrow t + 1$;
**until** $t > T_{\max}$;
**return** $\boldsymbol{W}$.

To train the network, we adopt the cross-entropy loss function to measure the discrepancy between the predicted probabilities and the true labels. Specifically, define the index set of labeled samples as $\mathcal{D}_L = \{p_1, p_2, \ldots, p_N\}$, where $N$ represents the number of labeled samples. For each pixel $p_t$, the one-hot encoding of its ground truth label is denoted as

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$\boldsymbol{L}(p_t)$, and the predicted probability is $\boldsymbol{Y}(p_t)$. Then, the cross-entropy loss function is used to optimize the PHHF, which is defined as

$$\mathcal{L}_{\mathrm{PHHF}}(\boldsymbol{W}) = -\frac{1}{|\mathcal{D}_L|}\sum_{t=1}^{N}\sum_{c=1}^{C}\boldsymbol{L}_c(p_t)\log \boldsymbol{Y}_c(p_t) \tag{19}$$

where $\boldsymbol{W}$ refers to the set containing all trainable parameters in the PHHF, $\boldsymbol{Y}_c(p_t)$ denotes the predicted probability of the labeled pixel $p_t$ belonging to class $c$, $\boldsymbol{L}_c(p_t)$ represents the $c$th dimension of one-hot ground truth label (1 if this pixel belongs to class $c$, and 0 otherwise), and $C$ denotes the total number of land cover classes in the HSI dataset.

## III. Experiment

### A. Experiment Settings

*1) Datasets:* In this study, experiments are conducted using three benchmark HSI datasets: the Indian Pines dataset, the Xuzhou dataset, and the Kansas dataset.

The Indian Pines dataset was captured by the AVIRIS sensor over an agricultural test site in North-western Indiana. The image size is 145 × 145 with a spatial resolution of 20 m/pixel. It contains 200 spectral reflectance bands in the wavelength range of 0.4–2.5 $\mu$m. This dataset includes 16 distinct land cover classes.

The Kansas dataset [45] was acquired by the visible-SWIR AHSI onboard the GF-5 satellite over Kansas, USA. The scene has a size of 650 × 340 pixels with a spatial resolution of 30 m/pixel. It consists of 308 spectral bands to characterize the solar reflective regime from 400 to 2500 nm. This dataset includes seven land cover classes.

The Xuzhou dataset [46] was collected over a peri-urban site in Xuzhou, China. It consists of 500 × 260 pixels with a very high spatial resolution of 0.73 m/pixel. After removing the noisy bands ranging from 415 to 2508 nm, 436 spectral bands are utilized in the experiments. The scene contains nine challenging land cover categories.

*2) Comparison Methods:* The PHHF is compared against several state-of-the-art HSI classification models to verify its effectiveness, including CNN-based methods, i.e., the cascaded spatial cross-attention network (CSCANet) [47] and the composite neighbor-aware convolutional metric network (CNCMN) [48], GCN-based methods, i.e., GCN, CEGCN [29], the multilevel superpixel structured graph U-Nets (MSSGU) [32], the AMGCFN [30], the two-branch deeper GCN (TBDGCN) [49], and the self-supervised graph masked autoencoders (SGMAEs) [35], and also HGCN-based methods, i.e., the hypergraph convolutional network with multiple hyperedges fusion (HGCN-MHF) [41] and the CNN-enhanced hypergraph attention (CEHGAT) [42]. All compared methods are implemented using the hyperparameter settings in their respective original publications.

*3) Parameter Setting:* In our experiments, the hyperparameter $L$ is fixed at 5 for all three datasets. The learning rate $\eta$ is set to 0.0005, and the number of training iterations $T_{\max}$ is set to 300. All experiments are carried out on a desktop workstation equipped with an Intel Xeon W-2245 CPU and an NVIDIA GeForce RTX A4000 GPU. The training, validation, and testing samples of the three datasets are shown in Tables I–III, respectively. Specifically, for the Kansas and Xuzhou datasets, we select ten labeled samples per class for training and validation, and the remaining samples are used for testing. For the Indian Pines dataset, due to the limited number of labeled samples in classes 7 and 9, we select five labeled samples from each class for training and validation. For the remaining classes, we select ten labeled samples per class for training and validation.

TABLE I
Numbers of Training, Valid, and Testing Samples for Indian Pines Dataset

| No. | Color | Land cover type | Train | Valid | Test |
|---|---|---|---|---|---|
| 1 | | Alfalfa | 10 | 10 | 26 |
| 2 | | Corn-notill | 10 | 10 | 1408 |
| 3 | | Corn-mintill | 10 | 10 | 810 |
| 4 | | Corn | 10 | 10 | 217 |
| 5 | | Pasture | 10 | 10 | 463 |
| 6 | | Grass-trees | 10 | 10 | 710 |
| 7 | | Pasture-mowed | 5 | 5 | 18 |
| 8 | | Hay-windrowed | 10 | 10 | 458 |
| 9 | | Oats | 5 | 5 | 10 |
| 10 | | Soybean-notill | 10 | 10 | 952 |
| 11 | | Soybean-mintill | 10 | 10 | 2435 |
| 12 | | Soybean-clean | 10 | 10 | 573 |
| 13 | | Wheat | 10 | 10 | 185 |
| 14 | | Woods | 10 | 10 | 1245 |
| 15 | | Building-Grass | 10 | 10 | 366 |
| 16 | | Stone-Steel Towers | 10 | 10 | 73 |
| Total | | | 150 | 150 | 9949 |

TABLE II
Numbers of Training, Valid, and Testing Samples for Kansas Dataset

| No. | Color | Land cover type | Train | Valid | Test |
|---|---|---|---|---|---|
| 1 | | Crop | 10 | 10 | 5183 |
| 2 | | Meadow | 10 | 10 | 5448 |
| 3 | | Business Area | 10 | 10 | 4675 |
| 4 | | Bare Soil | 10 | 10 | 12068 |
| 5 | | Residential Area | 10 | 10 | 8215 |
| 6 | | Water | 10 | 10 | 1663 |
| 7 | | Road | 10 | 10 | 4561 |
| Total | | | 70 | 70 | 41813 |

TABLE III
Numbers of Training, Valid, and Testing Samples for Xuzhou Dataset

| No. | Color | Land cover type | Train | Valid | Test |
|---|---|---|---|---|---|
| 1 | | Bareland-1 | 10 | 10 | 26376 |
| 2 | | Bareland-2 | 10 | 10 | 4007 |
| 3 | | crops-1 | 10 | 10 | 2763 |
| 4 | | crops-2 | 10 | 10 | 5194 |
| 5 | | Red-tiles | 10 | 10 | 13164 |
| 6 | | Trees | 10 | 10 | 2416 |
| 7 | | Coals | 10 | 10 | 6970 |
| 8 | | Cement | 10 | 10 | 4757 |
| 9 | | Lakes | 10 | 10 | 3050 |
| Total | | | 90 | 90 | 68697 |

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TABLE IV
CLASSIFICATION ACCURACIES OF THE COMPARED METHODS ON THE INDIAN PINES DATASET

| Class | Convolutional Neural Network | | Graph Neural Network | | | | | | Hypergraph Neural Network | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | CNCMN | CSCANet | GCN | CEGCN | MSSGU | AMGCFN | TBDGCN | SGMAE | HGCN-MHF | CEHGAT | PHHF |
| 1 | **100.00±0.00** | **100.00±0.00** | 97.92±1.84 | 97.60±3.15 | **100.00±0.00** | **100.00±0.00** | 97.36±3.64 | 36.96±21.13 | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** |
| 2 | 80.03±11.68 | 70.70±10.34 | 73.87±12.50 | 64.70±8.97 | 77.94±13.87 | 81.89±5.18 | 69.75±11.17 | 80.97±6.45 | 72.46±14.17 | 81.34±12.82 | **82.33±8.47** |
| 3 | 86.05±8.14 | 82.81±5.00 | 70.87±11.65 | 62.89±16.11 | 82.59±7.21 | 81.82±15.95 | 86.84±7.05 | 83.72±15.61 | 82.96±13.03 | 80.57±10.83 | **88.47±4.46** |
| 4 | 96.43±4.88 | 95.05±3.86 | 93.39±5.12 | 97.04±2.94 | 93.72±5.22 | 90.67±11.10 | 95.68±6.60 | 61.81±12.12 | 90.06±10.95 | 97.51±1.62 | **99.30±1.50** |
| 5 | 83.91±7.00 | 86.93±3.63 | 82.37±7.42 | 89.22±3.65 | 85.50±5.56 | 81.21±2.63 | 85.18±10.48 | **90.00±4.42** | 85.28±6.22 | 82.46±6.39 | 85.81±7.62 |
| 6 | 97.25±2.63 | 94.65±3.01 | 91.80±7.80 | **99.09±1.03** | 93.94±6.66 | 95.67±2.92 | 95.14±6.25 | 96.95±2.86 | 97.99±1.11 | 97.13±1.08 | 98.30±1.09 |
| 7 | **100.00±0.00** | 97.22±5.56 | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | 85.71±11.90 | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** |
| 8 | 99.97±0.08 | 99.13±1.35 | 99.97±0.07 | 99.84±0.37 | **100.00±0.00** | 93.83±10.40 | 99.96±0.09 | 98.37±2.33 | 99.61±1.17 | 99.83±0.39 | 99.96±0.09 |
| 9 | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | 98.13±7.50 | 64.50±14.42 | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** |
| 10 | 79.85±6.40 | 77.63±4.62 | 77.48±6.05 | 79.48±6.71 | 82.48±8.14 | 81.75±6.38 | 84.89±7.81 | 78.78±5.68 | 81.86±11.12 | 78.49±9.59 | **90.40±6.04** |
| 11 | 81.23±6.34 | 82.26±3.88 | 75.45±7.64 | 79.59±6.07 | 76.81±16.77 | 69.67±4.16 | 81.87±8.35 | 78.92±2.15 | 73.85±11.23 | 76.90±8.25 | **82.50±3.26** |
| 12 | 88.63±5.27 | 67.28±9.57 | 80.86±16.59 | 78.79±9.03 | **91.34±6.58** | 74.83±15.56 | 83.73±10.93 | 69.43±14.49 | 75.71±16.19 | 74.35±18.87 | 90.43±7.97 |
| 13 | 99.59±0.56 | 98.38±1.08 | 97.94±2.29 | 99.78±0.37 | 99.39±1.10 | 98.24±1.61 | 99.53±1.18 | 98.29±1.74 | 99.45±0.44 | 99.35±0.24 | **100.00±0.00** |
| 14 | 97.42±1.94 | 97.95±1.48 | 93.70±4.18 | 96.05±4.64 | **99.77±0.36** | 95.32±3.55 | 96.44±4.18 | 98.31±1.24 | 97.08±2.78 | 96.63±5.32 | 99.49±1.14 |
| 15 | **99.73±0.33** | 91.73±5.64 | 86.09±10.38 | 83.64±6.92 | 93.00±8.56 | 96.79±2.60 | 98.43±3.86 | 85.21±10.15 | 91.14±9.90 | 88.47±12.64 | 94.21±7.85 |
| 16 | 99.49±1.02 | 87.67±0.92 | 96.68±8.32 | 99.83±0.42 | **99.84±0.48** | 96.58±2.85 | 98.97±1.37 | 91.61±15.64 | 99.45±0.96 | 96.16±6.53 | 99.56±1.09 |
| OA(%) | 87.49±2.54 | 84.26±2.15 | 81.81±4.04 | 82.14±1.65 | 86.07±3.08 | 82.91±0.49 | 86.17±3.60 | 86.68±3.91 | 83.39±4.32 | 84.62±2.69 | **89.77±1.63** |
| AA(%) | 93.10±1.30 | 89.34±1.23 | 88.65±2.73 | 89.22±1.16 | 92.27±1.06 | 89.89±1.22 | 91.99±2.43 | 81.85±4.89 | 90.43±2.70 | 90.57±1.72 | **94.36±1.04** |
| $\kappa \times 100$ | 85.80±2.85 | 82.06±2.35 | 79.52±4.49 | 79.64±1.88 | 84.16±3.40 | 80.63±0.54 | 84.33±3.65 | 84.77±4.53 | 81.08±4.88 | 82.45±2.95 | **88.39±1.85** |

*4) Indices:* To quantitatively assess the classification performance, we adopt several widely used evaluation measures, including overall accuracy (OA), average accuracy (AA), and the Kappa coefficient ($\kappa$) [50]. In general, higher values reflect better classification accuracies. All experimental results are presented as the mean accuracy obtained from ten trials.

## B. Classification Performance

*1) Indian Pines Dataset:* To assess the effectiveness of PHHF, we first conduct experiments on the Indian Pines dataset. A summary of the results achieved by different methods is provided in Table IV, where the best performance in each row is marked in bold. Specifically, GCN yields poor results with an OA of less than 82%. CSCANet shows superior performance compared with GCN, achieving higher OA due to the incorporation of a cascaded spatial cross-attention module and a two-branch feature separation structure. CNCMN further improves classification accuracy by learning each target's representation from its composite neighbors. CEGCN achieves relatively better results than GCN because it incorporates CNN-based pixel-level feature extraction to enhance the capture of local details. MSSGU demonstrates significant improvement in OA and $\kappa$, and achieves the best performance across multiple categories, which confirms the effectiveness of the graph U-Net architecture. TBDGCN performs excellently in terms of AA, which confirms the effectiveness of the two-branch deeper GCN. SGMAEs produce better classification performance in terms of OA and $\kappa$. Compared with GCN and CEGCN, HGCN-MHF produces better classification performance in terms of OA, AA, and $\kappa$. A possible reason is that HGCN-MHF leverages the hypergraph to model the spatial information of HSIs and could capture high-order interactions within HSIs well. CEHGAT achieves relatively better results than CEGCN because it incorporates a hypergraph attention network branch to learn more representative node embeddings. Notably, PHHF outperforms all baseline approaches across all evaluation metrics, and also exhibits optimal accuracy in nine out of 16 classes. Moreover, PHHF demonstrates remarkable robustness to class imbalance, achieving 2.09%–12.51% improvement in AA over the compared methods while maintaining excellent performance on classes with limited labeled samples. In summary, the PHHF achieves state-of-the-art performance on the Indian Pines dataset.

Fig. 7 illustrates the classification maps generated by various methods. The ground truth is displayed first for convenience. CSCANet fails to preserve spatial structures effectively. CNCMN exhibits noticeable classification errors in many fine details. The classification map generated by GCN exhibits significantly degraded visual quality. Some local details are lost in the classification map of MSSGU, while AMGCFN could not preserve the spatial structures well. The classification map produced by TBDGCN suffers from noticeable noise artifacts, particularly in spatially homogeneous regions. SGMAEs fails to retain local details, while CEHGAT struggles to preserve the underlying spatial structures effectively. As shown in Fig. 7, it is obvious that our proposed PHHF produces a classification map that is most consistent with the ground truth compared with those of other methods.

*2) Kansas Dataset:* Table V summarizes the classification results on the Kansas Dataset. As shown in Table V, the OA of CSCANet exceeds 80%. CNCMN demonstrates outstanding performance, particularly achieving the highest classification accuracy on Class 5. GCN exhibits poor performance across all evaluation metrics. AMGCFN shows superior performance compared with GCN, achieving higher OA due to the incorporation of multihop graph structures and CNN-based pixel-level feature extraction. Moreover, MSSGU further improves classification accuracy by leveraging a multiscale framework. TBDGCN outperforms GCN and AMGCFN. SGMAEs achieves further improvement in OA. A possible reason is that it innovatively employs graph masked autoencoders to achieve self-supervised label-free feature extraction. CEHGAT exhibits the highest performance among all methods except PHHF, and achieves the highest accuracy of 93.03% on Class 6. Notably, PHHF achieves the best OA, AA, and $\kappa$ compared with other methods, and also outperforms the baselines in three out of seven classes.

Fig. 8 illustrates the classification maps of the compared methods on the Kansas dataset. The classification map

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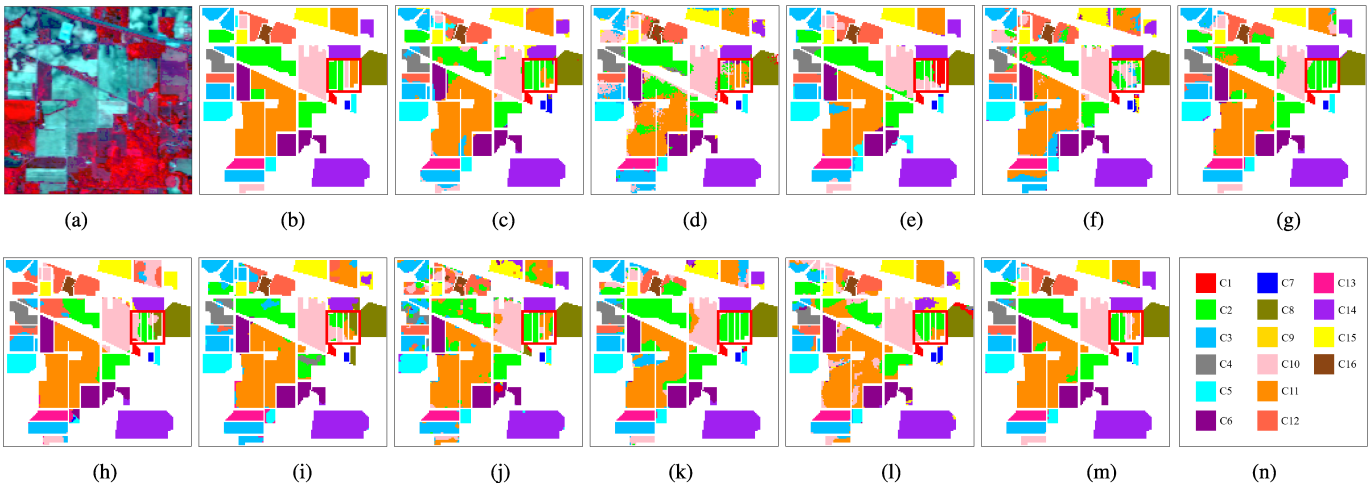



Fig. 7. Classification maps of the compared methods on the Indian Pines dataset. (a) False color image. (b) Ground truth. (c) CNCMN. (d) CSCANet. (e) GCN. (f) CEGCN. (g) MSSGU. (h) AMGCFN. (i) TBDGCN. (j) SGMAEs. (k) HGCN-MHF. (l) CEHGAT. (m) PHHF. (n) Colors ground.

TABLE V

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS ON THE KANSAS DATASET

| Class | Convolutional Neural Network | | Graph Neural Network | | | | | | Hypergraph Neural Network | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | CNCMN | CSCANet | GCN | CEGCN | MSSGU | AMGCFN | TBDGCN | SGMAE | HGCN-MHF | CEHGAT | PHHF |
| 1 | 92.68±6.87 | 93.99±3.64 | 81.62±5.24 | 90.89±6.12 | 93.53±5.04 | 88.71±9.95 | 93.73±5.56 | 91.35±9.07 | **97.89±1.89** | 97.66±1.23 | 96.61±2.74 |
| 2 | 95.92±2.14 | 93.63±3.20 | 87.01±4.05 | 97.43±1.31 | 95.33±7.56 | 97.51±1.20 | 97.37±2.01 | 97.11±1.88 | 97.37±2.49 | 97.65±1.23 | **98.67±1.61** |
| 3 | 77.43±18.76 | 69.32±6.56 | 68.02±12.80 | 81.38±7.07 | 84.72±7.85 | 80.48±8.65 | 78.99±5.20 | 76.51±5.28 | 86.84±9.42 | 83.51±10.98 | **89.90±4.46** |
| 4 | 86.60±5.05 | 84.23±7.09 | 85.65±7.74 | 87.33±8.52 | 89.12±8.54 | 86.38±8.09 | 88.35±5.82 | 89.94±4.62 | 82.25±5.66 | 84.58±3.20 | **90.39±3.39** |
| 5 | **95.41±4.46** | 86.19±6.70 | 80.04±9.77 | 89.98±6.09 | 90.21±5.31 | 84.04±14.11 | 86.64±5.97 | 91.40±3.26 | 87.93±2.98 | 92.92±7.14 | 93.36±4.17 |
| 6 | 81.44±7.37 | 77.16±5.73 | 58.35±10.67 | 83.29±8.00 | 73.29±12.45 | 81.89±9.89 | 82.66±7.08 | 69.47±4.17 | 91.98±10.96 | **93.03±2.86** | 81.86±4.33 |
| 7 | 48.42±8.96 | 38.36±9.53 | 38.84±8.95 | 46.72±5.69 | 47.07±5.83 | **51.83±7.32** | 46.56±10.20 | 47.52±5.14 | 49.95±7.94 | 46.34±10.24 | 46.71±8.44 |
| OA(%) | 84.90±2.47 | 80.10±3.53 | 76.06±1.51 | 84.35±2.39 | 84.98±2.27 | 83.97±3.90 | 84.03±3.76 | 84.38±2.16 | 84.65±1.45 | 85.59±2.58 | **87.67±1.07** |
| AA(%) | 82.56±2.27 | 77.55±2.32 | 71.36±2.16 | 82.43±2.01 | 81.89±1.72 | 81.55±4.63 | 82.04±1.80 | 80.47±1.94 | 84.89±1.17 | 85.10±2.27 | **85.36±1.16** |
| $\kappa \times 100$ | 81.66±2.92 | 75.86±3.99 | 70.90±1.74 | 80.99±2.79 | 81.75±2.63 | 79.50±4.71 | 80.63±4.26 | 80.92±2.62 | 81.50±1.66 | 82.56±3.07 | **85.00±1.28** |

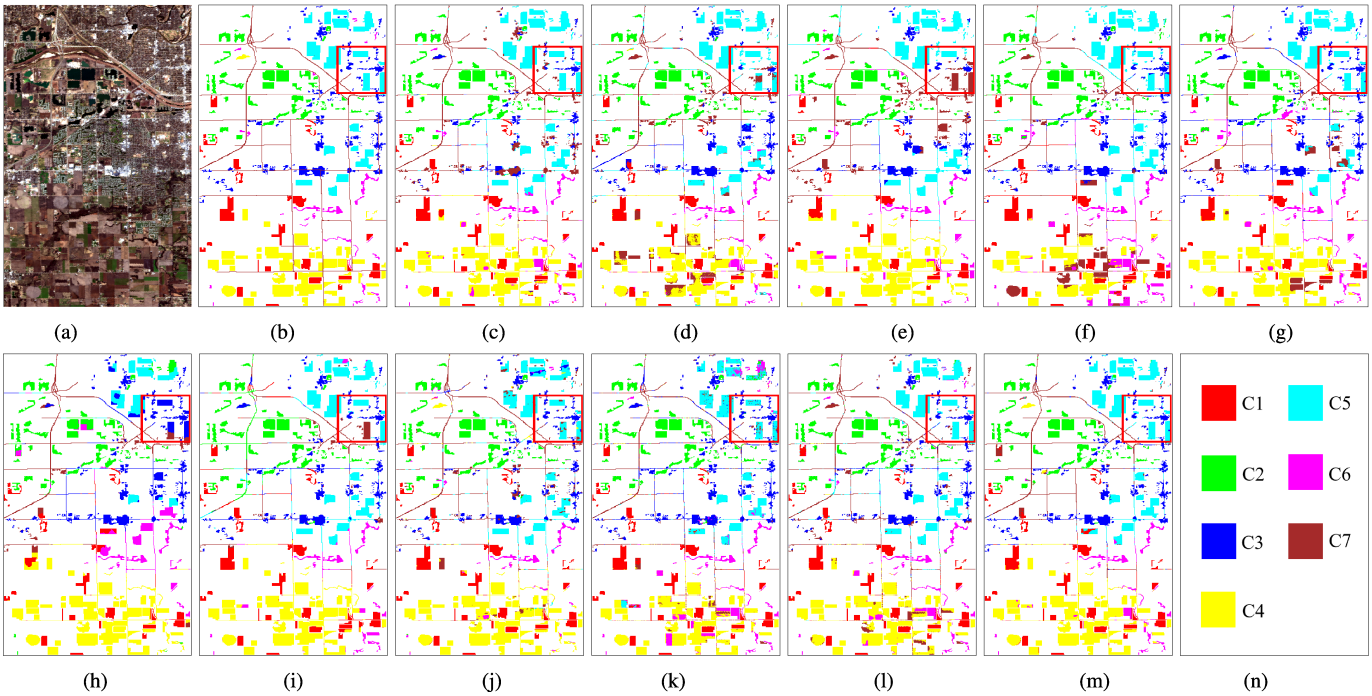



Fig. 8. Classification maps of the compared methods on the Kansas dataset. (a) False color image. (b) Ground truth. (c) CNCMN. (d) CSCANet. (e) GCN. (f) CEGCN. (g) MSSGU. (h) AMGCFN. (i) TBDGCN. (j) SGMAEs. (k) HGCC-MHF. (l) CEHGAT. (m) PHHF. (n) Colors ground.

generated by GCN exhibits a large number of noisy pixels. The classification map of CSCANet exhibits degraded visual quality, with noticeable noise artifacts in homogeneous areas. In contrast, CNCMN shows clear improvements in the visual classification map. MSSGU demonstrates considerably fewer misclassifications compared with GCN, as it employs

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TABLE VI
CLASSIFICATION ACCURACIES OF THE COMPARED METHODS ON THE XUZHOU DATASET

| Class | Convolutional Neural Network | | Graph Neural Network | | | | | | Hypergraph Neural Network | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | CNCMN | CSCANet | GCN | CEGCN | MSSGU | AMGCFN | TBDGCN | SGMAE | HGCN-MHF | CEHGAT | PHHF |
| 1 | 89.60±2.71 | 92.10±2.31 | 91.06±2.75 | 89.76±1.78 | 93.22±2.98 | 90.72±0.18 | 90.31±3.18 | 95.43±1.26 | **95.17±1.47** | 93.69±1.70 | 93.64±1.26 |
| 2 | 99.45±0.19 | 98.28±0.67 | 98.72±1.00 | 99.39±0.20 | 99.51±0.31 | 98.84±0.86 | 99.20±0.36 | 99.03±1.17 | 98.43±1.67 | 98.62±0.82 | **99.56±0.22** |
| 3 | 99.02±1.15 | 95.58±4.95 | 98.47±2.39 | 99.73±0.45 | 97.48±2.65 | 99.12±0.93 | **99.82±0.23** | 82.03±4.86 | 95.62±3.95 | 96.31±4.29 | 99.16±0.51 |
| 4 | 92.60±3.72 | 90.90±3.69 | 86.45±9.14 | 92.23±6.37 | 86.54±10.51 | 89.63±5.83 | 91.97±6.10 | 92.16±3.49 | 88.02±6.63 | 89.04±7.72 | **92.65±5.08** |
| 5 | 98.38±2.76 | 96.48±1.86 | 97.13±5.16 | **98.50±0.73** | 96.06±2.09 | 97.92±1.40 | 96.98±2.53 | 95.78±2.08 | 90.52±5.68 | 91.64±1.99 | 97.17±1.24 |
| 6 | 92.28±8.59 | 95.37±6.08 | 98.42±1.20 | **99.12±0.98** | 97.73±5.04 | 98.09±2.93 | 95.69±9.77 | 82.47±4.38 | 96.29±7.12 | 91.44±5.64 | 98.76±2.47 |
| 7 | 99.49±0.64 | 97.92±1.96 | 97.59±2.34 | 99.28±0.52 | 98.46±2.31 | 99.32±0.60 | 98.42±2.00 | 88.56±3.21 | 96.18±3.20 | 98.60±0.81 | **99.54±0.71** |
| 8 | 99.90±0.22 | 99.17±0.72 | 99.60±0.33 | 99.97±0.04 | **100.00±0.00** | 99.95±0.14 | 99.97±0.05 | 98.20±1.65 | 99.96±0.05 | 99.60±0.30 | **100.00±0.00** |
| 9 | 98.30±1.63 | 97.74±0.70 | 83.39±11.26 | 97.89±1.11 | 94.41±5.33 | 95.24±4.82 | 97.55±2.21 | **98.76±4.01** | 96.87±3.61 | 95.62±1.85 | 96.72±1.67 |
| OA(%) | 94.86±1.48 | 94.79±1.50 | 93.79±1.31 | 94.95±0.89 | 95.01±1.22 | 94.80±0.78 | 94.69±1.34 | 94.23±0.52 | 94.50±1.57 | 94.25±1.29 | **96.16±0.46** |
| AA(%) | 96.85±1.16 | 95.95±1.28 | 94.54±1.43 | 97.35±0.64 | 95.93±1.66 | 96.54±0.87 | 96.76±1.29 | 92.67±0.79 | 95.23±1.63 | 94.95±1.54 | **97.47±0.59** |
| $\kappa \times 100$ | 93.56±1.83 | 93.45±1.85 | 92.20±1.62 | 93.68±1.10 | 93.70±1.56 | 93.48±0.97 | 93.34±1.65 | 92.69±0.65 | 93.05±1.98 | 92.76±1.61 | **95.17±0.57** |

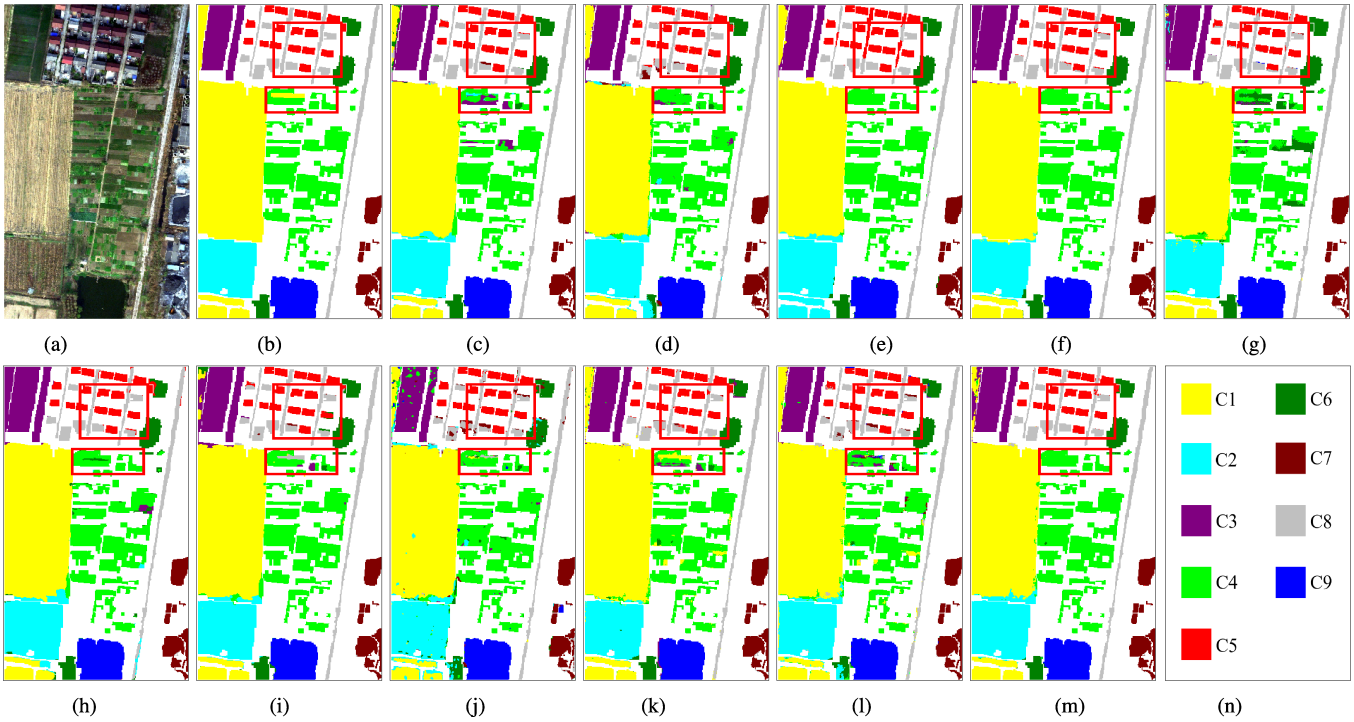



Fig. 9. Classification maps of the compared methods on the Xuzhou dataset. (a) False color image. (b) Ground truth. (c) CNCMN. (d) CSCANet. (e) GCN. (f) CEGCN. (g) MSSGU. (h) AMGCFN. (i) TBDGCN. (j) SGMAEs. (k) HGCN-MHF. (l) CEHGAT. (m) PHHF. (n) Colors ground.

a multiscale framework and incorporates CNN for pixel-level feature extraction. SGMAEs demonstrates relatively superior classification accuracy in the region highlighted by the red box. CEHGAT demonstrates considerably fewer misclassifications than GCN. Compared with these baselines, PHHF reduces noise and exhibits better visual performance on the Kansas dataset.

*3) Xuzhou Dataset:* Table VI summarizes the classification results of various methods on the Xuzhou Dataset. Specifically, the CSCANet demonstrates satisfactory performance. CNCMN achieves an AA of 96.85% and attains an accuracy of 99.49% on Class 7. The CEGCN outperforms the GCN in terms of OA, AA, and $\kappa$. This improvement is due to the incorporation of CNN, which improves its capability to capture local details. MSSGU surpasses the CEGCN across many metrics. This is attributed to its adoption of a multiscale framework, which effectively handles spatial information at different scales. SGMAEs further improve the OA and $\kappa$, and achieve the highest accuracy of 98.76% on Class 9. Notably, PHHF outperforms all baseline approaches across all evaluation metrics, owing to its hybrid-order hypergraph strategy within a multiscale progressive framework, which effectively models complex spectral–spatial relationships. PHHF also attains the highest accuracy in four out of nine classes.

Fig. 9 illustrates the classification maps generated by various methods on the Xuzhou Dataset. CSCANet and CNCMN achieve relatively good visual performance in the classification maps. However, some misclassifications still occur in local details. Compared with GCN, CEHGAT substantially reduces noisy pixels in heterogeneous areas. Notably, PHHF further suppresses salt-and-pepper noise, producing the classification map that is much closer to the ground truth than those generated by other methods. The above discussions demonstrate the effectiveness of PHHF on the Xuzhou Dataset.

*4) Classification Under Different Numbers of Training Samples:* This section investigates the effect of varying the

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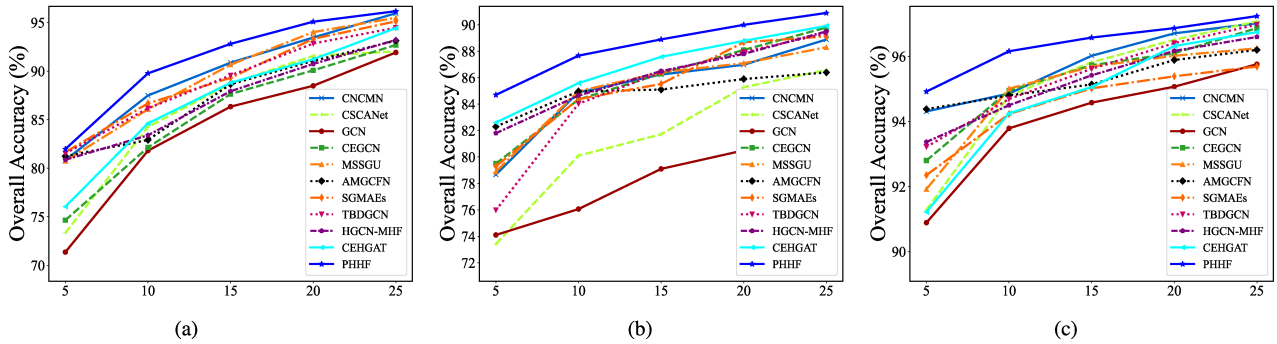



Fig. 10. Classification results of the compared methods with different numbers of training samples on three datasets. (a) Indian Pines. (b) Kansas. (c) Xuzhou.

TABLE VII
COMPUTATIONAL COMPLEXITY ON THREE DATASETS

| Methods | Indian Pines | | | Kansas | | | XuZhou | | |
|---|---|---|---|---|---|---|---|---|---|
| | Training time | Testing time | Parameters | Training time | Testing time | Parameters | Training time | Testing time | Parameters |
| GCN | 1.81 | 0.0012 | 0.13M | 18.06 | 0.0023 | 0.14M | 9.65 | 0.0040 | 0.15M |
| CEGCN | 4.70 | 0.0032 | 0.17M | 47.03 | 0.0060 | 0.18M | 25.14 | 0.0107 | 0.19M |
| MSSGU | 8.85 | 0.0054 | 0.14M | 88.53 | 0.0103 | 0.15M | 47.32 | 0.0183 | 0.16M |
| AMGCFN | 11.61 | 0.0043 | 0.37M | 116.12 | 0.0080 | 0.39M | 62.07 | 0.0143 | 0.43M |
| TBDGCN | 7.31 | 0.0036 | 0.49M | 76.53 | 0.0071 | 0.50M | 42.35 | 0.0104 | 0.52M |
| SGMAEs | 8.77 | 0.0056 | 0.40M | 98.29 | 0.0120 | 0.40M | 41.68 | 0.0135 | 0.40M |
| HGCN-MHF | 10.40 | 0.0094 | 0.28M | 104.03 | 0.0177 | 0.31M | 55.61 | 0.0315 | 0.33M |
| CEHGAT | 8.74 | 0.0044 | 0.21M | 91.22 | 0.0135 | 0.24M | 83.63 | 0.0180 | 0.27M |
| PHHF | 7.29 | 0.0028 | 0.13M | 72.90 | 0.0052 | 0.14M | 38.97 | 0.0093 | 0.16M |

number of training samples on classification accuracy. As illustrated in Fig. 10, several existing methods struggle to achieve satisfactory performance when only a small number of labeled samples are available. Specifically, GCN, CEGCN, and CEHGAT perform relatively poorly under the five-sample-per-class setting on the Indian Pines datasets, but show substantial performance improvement when the number of training samples is increased to ten per class. On the Kansas dataset, CSCANet and TBDGCN exhibit relatively poor performance when the number of training samples per class is limited. On the Xuzhou dataset, GCN, CEHGAT, and MSSGU achieve subpar results when trained with only five samples per class. In contrast, the proposed PHHF consistently maintains high classification performance across all evaluated sample sizes. Notably, PHHF significantly outperforms the comparison methods, especially in scenarios with extremely limited labeled data.

*5) Computational Complexity:* The computational analysis in Table VII provides the following key observations. Overall, the basic GCN exhibits the shortest running time among all methods due to its structural simplicity. For instance, on the Indian Pines dataset, GCN requires only 1.81 s for training; however, this efficiency comes at the cost of relatively poor classification accuracy. AMGCFN and HGCN-MHF require longer training times, which can be attributed to the multiscale CNN modules that demand extensive computations. On the Kansas dataset, AMGCFN requires 116.12 s for training, while HGCN-MHF takes 104.03 s. TBDGCN has a relatively large number of parameters (e.g., 0.52M on the Xuzhou dataset) due to its deep two-branch architecture. Our proposed PHHF requires longer training time than CEGCN and GCN, primarily due to the progressive structure that increases computational overhead. However, PHHF exhibits clear advantages over other compared methods. In terms of testing time, PHHF is the fastest among all methods except GCN, highlighting its competitive efficiency. For example, on the Indian Pines dataset, PHHF requires only 0.0028 s for testing, outperforming most methods. Moreover, PHHF maintains the lowest parameter count among all methods except the basic GCN, benefiting from the lightweight design of HetConv. On the Kansas dataset, PHHF contains only 0.14M parameters, markedly fewer than AMGCFN (0.39M) and HGCN-MHF (0.31M). In summary, although PHHF requires longer training time than simpler GCN-based methods, its superior performance makes it a robust choice for HSI classification.

*6) Parameter Analysis:* This section aims to investigate the influence of the hyperparameter $L$ on classification performance. Specifically, we use the grid search method to find the optimal hyperparameter $L$ from $\{2, 3, \ldots, 6\}$ for each dataset. As shown in Fig. 11, as $L$ increases, the classification accuracy of the PHHF initially improves and then plateaus after five layers for the Indian Pines and Kansas datasets. For the Xuzhou dataset, the highest accuracy is achieved with four layers, with further increases in depth yielding diminishing returns. This suggests that while appropriately increasing the framework depth is beneficial, excessive depth does not necessarily improve performance. Moreover, a larger number of layers increases computational cost. In practice, it is important to select an appropriate framework depth to balance the effectiveness and efficiency of the PHHF.

## C. Ablation Study

To comprehensively evaluate the effectiveness of our proposed PHHF, we conducted a series of ablation studies.

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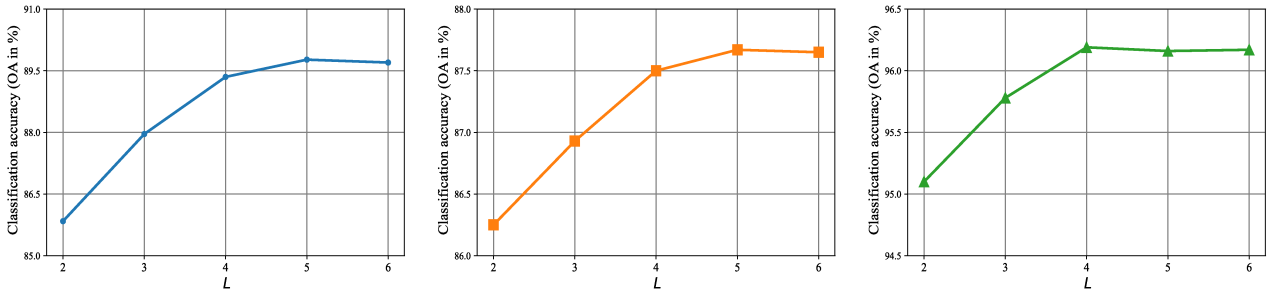



Fig. 11. Analysis of the hyperparameter $L$ on three datasets. (a) Indian Pines. (b) Kansas. (c) Xuzhou.

TABLE VIII
RESULTS OF THE ABLATION STUDY ON THE INDIAN PINES, KANSAS, AND XUZHOU DATASETS

| | Module | | Indian Pines | | | Kansas | | | XuZhou | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | HOHG | PHS | OA(%) | AA(%) | $\kappa \times 100$ | OA(%) | AA(%) | $\kappa \times 100$ | OA(%) | AA(%) | $\kappa \times 100$ |
| $PHHF_{base}$ | × | × | 78.99 | 88.80 | 76.39 | 82.24 | 83.67 | 78.85 | 91.84 | 92.19 | 91.23 |
| $PHHF_{H}$ | ✓ | × | 82.57 | 90.41 | 80.18 | 83.61 | 84.83 | 80.30 | 93.69 | 94.57 | 92.03 |
| $PHHF_{P}$ | × | ✓ | 85.33 | 91.45 | 83.33 | 86.26 | 84.51 | 83.34 | 94.74 | 94.97 | 93.34 |
| PHHF | ✓ | ✓ | **89.77** | **94.36** | **88.39** | **87.67** | **85.36** | **85.00** | **96.16** | **97.47** | **95.17** |

Specifically, we constructed three variant models in the ablation study. The first variant, denoted as $PHHF_{base}$, employs a conventional hypergraph structure combined with the HetConv to extract features. The second variant, denoted as $PHHF_H$, integrates the proposed hybrid-order hyperedge generation strategy (HOHG) into the $PHHF_{base}$. The third variant, denoted as $PHHF_P$, adopts the PHS based on the $PHHF_{base}$. The results of the ablation study are presented in Table VIII. The OA of $PHHF_{base}$ on the Indian Pines dataset falls below 80%, and its performance on the Kansas and Xuzhou datasets is also relatively modest. In contrast, $PHHF_H$ achieves significant improvements by incorporating the hybrid-order hyperedge generation strategy to extract complementary spatial information. Notably, on the Indian Pines dataset, it achieves substantial improvements of 3.58% in OA and 3.79% in $\kappa$. $PHHF_P$ also achieves notable improvements in classification accuracy across all datasets. Compared with $PHHF_{base}$, $PHHF_P$ achieves significant OA improvements of 6.34%, 4.02%, and 2.90% on the Indian Pines, Kansas, and Xuzhou datasets, respectively. Finally, the PHHF, which integrates both the hybrid-order hyperedge generation strategy and the PHS, achieves state-of-the-art performance across all evaluation metrics. In summary, the ablation study confirms the positive contributions of each proposed component and underscores their importance in enhancing overall classification accuracy.

## IV. CONCLUSION

In this article, we propose a novel PHHF for HSI classification. First, we build a progressive HGNN framework, which can fully exploit hierarchical spatial information within HSIs. Second, we design a hybrid-order hyperedge generation strategy, which can generate three types of hyperedges and effectively capture both low-order and high-order relationships between hyperspectral pixels. Finally, we utilize the HetConv to extract pixel-level features, which can complement the progressive hypergraph framework but only increase a limited computational cost. Experimental results demonstrate that PHHF achieves state-of-the-art performance across various evaluation metrics and visualization results.

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